

THE UNIVERSITY OF ZAMBIA

SCHOOL OF NATURAL AND APPLIED SCIENCES

DEPARTMENT OF COMPUTING AND INFORMATICS

FINAL YEAR PROJECT PROPOSAL (2026)

Project Title:

ConferenceCheck Mobile: Analytics, Meals and Notifications

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1. Introduction

The increasing scale and complexity of modern conferences require efficient systems to manage attendees, resources, and communication. Traditional conference management methods often rely on fragmented tools or manual processes, which lead to inefficiencies such as poor attendance tracking, mismanaged meal distribution, and delayed communication.

ConferenceCheck Mobile is a mobile-based system designed to centralize conference operations into a single platform. The system integrates analytics, digital meal voucher management, and real-time notifications to enhance both organiser efficiency and attendee experience.

This project is important because it addresses real-world challenges faced during conferences by improving transparency, reducing manual errors, and enabling real-time decision-making through data-driven insights.

1.1 Problem Statement

Many conferences lack an integrated system to manage operations effectively. Organisers face challenges such as:

- Lack of real-time attendance tracking
- Inefficient and sometimes fraudulent meal distribution systems
- Poor communication between organisers and attendees

These challenges reduce efficiency and negatively affect the overall conference experience.

1.2 Project Aim

To design and develop a mobile-based conference management system that provides real-time analytics, secure meal voucher redemption, and efficient push notification services.

1.3 Objectives

1. To develop a mobile application for conference management
2. To implement an analytics dashboard for tracking attendance
3. To design a secure QR-based meal voucher system
4. To integrate a push notification system for real-time updates

1.4 Scope of the Project

In Scope:

- Mobile application development using Flutter
- QR code-based meal voucher system
- Analytics dashboard
- Push notification system

Out of Scope:

- Payment systems
- Full web-based admin panel
- Integration with external ticketing platforms

2. Literature Overview

Mobile applications have significantly improved event management by providing real-time access to information and services. Research shows that mobile systems enhance coordination and user engagement during events.

QR code technology is widely used in secure verification systems such as ticketing and vouchers. It helps reduce fraud and improves operational efficiency.

Push notification services, such as those provided by cloud messaging platforms, enable real-time communication, ensuring that users receive important updates instantly.

However, most existing systems focus on individual functionalities rather than integrating analytics, meal management, and notifications into one platform. This project aims to provide a comprehensive solution.

3. Proposed Methodology

The project will follow an Agile (Scrum) development approach, allowing incremental development and continuous feedback.

The system will be developed in phases, starting with requirement analysis and system design, followed by implementation of core modules including authentication, meal voucher management, analytics, and notifications.

The system workflow will involve user registration, QR code generation for meal vouchers, scanning and validation of vouchers, real-time data updates, and notification delivery through cloud messaging services.

3.1 Tools and Technologies

Programming Language:

- Dart

Frameworks/Libraries:

- Flutter
- Provider / Riverpod
- fl_chart
- qr_flutter
- mobile_scanner
- Firebase Cloud Messaging

Database:

- MySQL / PostgreSQL

Backend:

- Laravel (REST API)

Development Tools:

- Android Studio / Visual Studio Code
- Git

3.2 Ethical and Legal Considerations

User data will be handled with strict confidentiality and security. The system will ensure:

- Protection of user data
- User consent before data collection
- Secure authentication and QR validation
- Prevention of unauthorized access

4. Expected Outcomes

The project will deliver:

- A functional mobile application
- A QR-based meal voucher system
- An analytics dashboard
- A push notification system
- Complete documentation and user manual

5. Project Timeline

Phase	Duration
Proposal	Week 1–2
Literature Review & Design	Week 3–5
Development	Week 6–10
Midway Evaluation	Week 11
Testing & Refinement	Week 12–13
Report Writing	Week 14–15
Final Submission	Week 16

6. Risk Analysis

Risk	Mitigation Strategy
QR code misuse	Backend validation and one-time use
Network issues	Offline support and retry mechanisms
Data loss	Regular backups
Time constraints	Agile sprint planning

7. References

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